

Assignment 3: Image Classification Using Pretrained CNN Architectures

Title:

Implementation and Performance Comparison of ResNet and GoogLeNet for Image Classification

Objective:

Implement and compare **ResNet** and **GoogLeNet/Inception** architectures using transfer learning on a standard image dataset.

Tasks:

1. Download and preprocess any dataset (CIFAR-10, Cats vs Dogs, or custom dataset).
2. Implement pretrained:
 - ResNet50
 - GoogLeNet / InceptionV3
3. Replace final classification layer according to number of classes.
4. Train both models for 10–20 epochs.
5. Compare:
 - Accuracy
 - Loss
 - Training time
 - Model size
6. Plot:
 - Accuracy vs Epoch
 - Loss vs Epoch
7. Write observations on why one model performed better.

Expected Deliverables:

1. Python code (TensorFlow / PyTorch)
2. Graphs
3. Confusion Matrix
4. Short report (3–5 pages)